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ORIGINAL RESEARCH

Blockchain enabled accurate contact tracing model (BEACTM) for future smart tourism industry

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Abstract
Effective contact tracing plays a vital role in controlling infectious diseases but becomes particularly challenging in highly mobile sectors such as tourism. This study presents the Blockchain-Enabled Accurate Contact Tracing Model (BEACTM), a secure and privacy-focused framework that harnesses blockchain's decentralized and immutable structure to enable precise data collection, verified information exchange, and automated alerts through smart contracts. User privacy is protected through cryptographic key management, digital identity mechanisms, and access control policies, ensuring that sensitive health information remains accessible only to authorized parties. Designed for interoperability with health and travel infrastructures, BEACTM supports cross-border implementation. Simulation results demonstrate significant improvements in accuracy, responsiveness, and trustworthiness over conventional systems, enabling safer and more resilient travel during pandemics. Beyond tourism, BEACTM is applicable to hospitals, public trans- port, and mass events, with future extensions integrating IoT and scalable blockchain architectures for broader public health protection."

Keywords Blockchain · COVID-19 · Smart contracts · Tourist · Contact tracing · Omicron · SARS-COV-2 · JN.1

1 Introduction

Contact tracing has been a cornerstone in controlling the spread of COVID-19 and its variants by identifying and managing individuals at risk of infection (Kumar et al. 2025). However, existing solutions often lack decentralization, transparency, security and reliability, which are crucial in high-mobility environments such as tourism. The tourism industry presents unique challenges for contact tracing due to the rapid cross-border movement of travelers, diverse infrastructures, and the need for swift, trustworthy data sharing (Kumar et al. 2025).

To address these gaps, this research proposes the Blockchain Enabled Accurate Contact Tracing Model (BEACTFM), a decentralized system that leverages blockchain's immutability and transparency to enhance trust, accountability, and secure health data management (Kumar et al. 2024). By integrating digital health passports, identity management, and smart contracts, BEACTFM automates exposure notifications and quarantine protocols while safeguard- ing user privacy. Unlike conventional centralized systems, it minimizes risks of data breaches and unauthorized access by relying on cryptographic security and distributed verification (Abdul-Raheem 2022).

The model is designed to interoperate with health and travel infrastructures ensuring seamless cross-border deployment. Simulation-based evaluation highlights its effectiveness in improving accuracy, responsiveness, and trust- worthiness, making it particularly suitable for the

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Table 1 Areas of impact in the tourism industry

References	Areas	Description
Vernekar (2020), Kumar et al. (2025)	Transparency	The traditional architecture that is currently in place lacks transparency.
Kumar et al. 2025; 2025	Settlement	Problems with financial transaction settlement faced by stakeholders
Kumar et al. (2025)	Loyalty	Not true ratings and reviews
Pincheira et al. (2020)	Fraud	With a centralized server, the architecture is not secure.
Kumar et al. (2025)	Identity	Identity theft is a concern
Vernekar (2020)	IoT	Despite the fact that the current architecture makes use of the IoT, it is not a full model.
Kumar et al. (2025)	Overbooking	Problems with the traffic flow are caused by the centralized character
Kumar et al. (2025)	Policy	Lack of confidence in the cost and control
Klaine et al. (2020)	No reliable contact finder app	To stop the COVID-19 epidemic from spreading further, a precise contact tracing app is urgently needed.

tourism sector where speed and reliability are critical for safe and resilient travel (Vernekar 2020).

1.1 Tourism

Tourism is one of the world's largest and most dynamic industries, compris ing multiple interrelated sectors such as transportation, accommodation, food and beverage services, travel agencies, tour operators, and a wide range of cul tural, natural, and entertainment attractions. Traditionally, the industry has operated through physical infrastructure and human-driven services, where travelers relied on face-to-face interactions, printed brochures, and centralized booking systems to plan their journeys. This conventional model not only contributed significantly to GDP, employment generation, and foreign. exchange earnings but also played a vital role in cultural exchange and regional development (Pahontu et al. 2020; Kumar et al. 2025).

However, despite its economic strength, traditional tourism faces persistent challenges. The reliance on manual processes often results in limited personalization of services, slower responsiveness to changing customer needs, and difficulties in integrating real-time data (Pincheira et al. 2020; Kumar et al. 2025). The global COVID-19 pandemic further exposed these vulnerabilities, revealing how unprepared the sector was to ensure traveler safety, maintain operational continuity, and implement rapid crisis management (Klaine et al. 2020). The absence of robust digital tools for health moni- toring, risk assessment, and contact tracing highlighted the urgent need for innovative, technology-driven frameworks(Ardito et al. 2019; Kumar et al. 2025). This shift has accelerated the demand for smart tourism solutions, where blockchain-enabled systems like BEACTM can enhance transparency, security, and trust while addressing the shortcomings of the conventional model (Bari et al. 2021; Kumar et al. 2025) (Table 1).

The tourism industry, driven by large-scale international and domestic travel, is highly susceptible to the rapid spread of infectious diseases. Constant mobility across borders, transient social interactions, and crowded environ ments such as airports, hotels, and attractions make identifying infection chains complex and time-sensitive. Figure 1 shows the Traditional Architecture of Tourism Services. Traditional contact tracing methods, which depend heavily on manual reporting and individual recall, have proven inadequate in such dynamic contexts, resulting in delays and gaps in detection (Mbunge 2020).

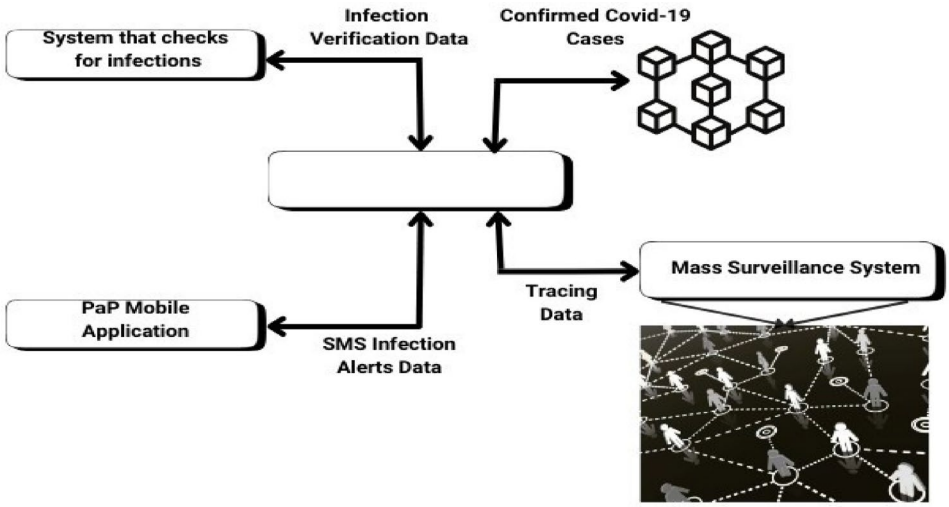


Fig. 1 Traditional architecture of tourism services

To address these challenges, technology-enabled solutions are essential. Blockchain, with its decentralized, transparent, and tamper-proof architecture, offers a promising foundation for building secure and efficient contact tracing models. By enabling real-time recording of traveler interactions while preserving privacy, blockchain-enhanced systems can significantly improve accuracy, responsiveness, and trust in pandemic management (Xu et al. 2020). Such innovations are particularly critical for the tourism sector, where restoring traveler confidence and ensuring regulatory compliance are prerequisites for sustainable recovery and safe border reopening (Önder and Treiblmaier 2017).

Traditional "test and trace" systems rely on iterative identification of con- tacts once an individual is diagnosed with an infectious disease. Although effective in principle, this process is burdened by administrative and logistical challenges, including delays in diagnostic testing, coordination difficulties, and secure data transfer between healthcare providers and authorities. These limitations often undermine the speed, accuracy, and privacy required for effective disease management (Pillai and Madhurya 2020). To overcome these gaps, blockchain-enabled solutions such as BEACTM provide a decentralized and tamper-proof framework for secure data exchange, real-time monitoring, and trustworthy communication among stakeholders. By ensuring transparency, accuracy, and privacy, BEACTM strengthens con- tact tracing in complex environments like tourism, where rapid response and traveler confidence are crucial (Iyengar et al. 2020; Abdul-Raheem 2022).

The tourism industry poses unique challenges for contact tracing because of its high mobility, cross-border interactions, and transient populations. Conventional systems, which depend on centralized databases, manual reporting, and.

personally identifiable information (PII), are often inefficient and vulnerable to delays, data breaches, and regulatory non-compliance (Tasatanattakool and Techapanupreeda 2018). These shortcomings not only slow down containment efforts but also reduce traveler willingness to share sensitive health data due to concerns about surveillance and misuse (Swati and Prasad 2018). Blockchain technology offers a transformative solution by providing a decentralized, transparent, and tamper-proof infrastructure for managing health and travel data (Seyito˘glu and Ivanov 2021). Its distributed architecture eliminates single points of failure, while cryptographic mechanisms safeguard privacy and ensure trust among stakeholders (Xu et al. 2021). By enabling secure, real-time, and privacy-preserving data exchange, blockchain can address the inherent limitations of traditional contact tracing and create the foundation for models like BEACTM, which are specifically designed to enhance safety and resilience in the tourism sector (Oksiiuk and Dmyrieva 2020).

1.2 Blockchain

The tourism industry, characterized by high mobility and cross-border movement, is highly vulnerable to the rapid spread of infectious diseases (Kumar et al. 2025; 2025). Traditional contact tracing systems, often centralized and dependent on manual reporting, face challenges such as delays, data inaccuracies, and privacy concerns. These limitations undermine traveler trust and hinder timely response, especially during large-scale health crises. To address these gaps, innovative, technology-driven frameworks are essential (Ricci et al. 2021).

Blockchain, a decentralized and tamper-proof distributed ledger, provides a secure infrastructure for recording and verifying transactions without reliance on a single authority (Xia et al. 2022). Its transparency and cryptographic safeguards foster trust among stakeholders while ensuring compliance with privacy regulations. In the tourism context, blockchain can record events such as check-ins, health status updates, and exposure alerts with integrity, anonymity, and accountability.

Through features like smart contracts, critical processes—including auto- mated exposure notifications or activation of isolation protocols—can be executed without human intervention, minimizing errors and delays. Addition- ally, blockchain's interoperability supports data sharing across borders and systems, a vital requirement for international travel (Xia et al. 2022).

By combining security, transparency, automation, and privacy preservation, blockchain (Hasan et al. 2021) offers a robust foundation for the Blockchain-Enabled Accurate Contact Tracing Model (BEACTM). This model is designed to overcome the inefficiencies of traditional methods and rebuild traveler confidence by enabling safe, resilient, and trustworthy tourism experiences (Kassab and Destefanis 2021; Girish Kumar et al. 2022).

Blockchain technology holds the potential to significantly transform the travel and tourism sector. With core features such as decentralization, transparency, immutability, and programmability, it enables the creation of innovative organizational models, workflows, and collaborative structures that span across traditionally siloed entities (Wen 2021).

In the digital age, e-tourism heavily relies on online payments processed through various types of payment gateways. However, these gateways can.

be vulnerable to cyberattacks, including identity theft, unauthorized access to digital wallets, and fraudulent payment processing. Centralized storage of payment data—often hosted on cloud servers—tends to be particularly

prob- lematic during peak usage times, leading to transaction failures and security breaches (Ozdemir et al. 2020).

Blockchain provides a decentralized alternative that promotes trust and efficient coordination among stakeholders such as banks, travel agencies, hotel chains, airlines, cruise lines, local transportation services, and restaurants (Narendran et al. 2017). By utilizing blockchain architecture, travelers can initiate secure transac- tions through a unified digital wallet connected to a cryptocurrency platform, streamlining communication with multiple service providers (Panchamia and Byrappa 2017).

One of the most profound advantages of blockchain in tourism is its abil- ity to reduce reliance on intermediaries. As digital travel platforms become increasingly common, the traditional supplier–customer relationship in the tourism industry is undergoing a fundamental shift (Sigala 2020). Blockchain facilitates direct interaction between service providers and consumers, thereby lowering costs and enhancing efficiency. Efficient contact tracing is essential for mitigat- ing the spread of infectious diseases, particularly in high-mobility sectors like tourism, where traditional methods often struggle to keep pace with rapid pop- ulation movement. Conventional contact tracing approaches typically rely on centralized data storage and manual reporting, which raise significant privacy concerns—including the risk of data breaches, unauthorized access, and misuse of personally identifiable information (Calvaresi et al. 2019). These vulnerabilities can undermine public trust and discourage participation, ultimately reducing the effectiveness of tracing efforts.

The Blockchain Enabled Accurate Contact Tracing Model (BEACTM) addresses these limitations by introducing a decentralized, secure, and privacy-preserving contact tracing system tailored for the tourism industry (Khare and Khare 2010). Rather than detailing each operational step, BEACTM emphasizes how blockchain's immutable ledger and cryptographic identifiers allow inter- actions to be logged securely and anonymously.

Smart contracts automate exposure notifications and isolation protocols, while access controls ensure that only authorized entities can view sensitive data. This model not only protects user privacy but also enhances trust, interoperability, and scalability across borders. By leveraging blockchain's core strengths, BEACTM offers a robust and future-ready solution for safe, resilient travel during global health crises (Dutta et al. 2020).

This project proposes a blockchain-based hotel booking system. It adopts a semi-structured research approach to gather insights from industry experts on how blockchain could reshape intermediary roles in the tourism sector (Bodkhe 2020). The findings indicate that blockchain is viewed as the most promising technology for removing traditional intermediaries from the booking process and prevent- ing new ones from emerging. Ultimately, this could lead to a more direct and efficient tourism ecosystem, free from unnecessary third-party intervention (Singh et al. 2022). By associating health records with cryptographic public and private key pairs rather than directly using personally identifiable data, BEACTM upholds the principles of data minimization and pseudonymization, thereby meeting key requirements of the GDPR and strengthening compliance with regulatory frameworks (Girish Kumar et al. 2023).

The Blockchain-Enabled Accurate Contact Tracing Model (BEACTM) introduces an innovative approach to overcome the shortcomings of traditional contact tracing methods by offering a decentralized, secure, and privacy-focused system tailored to the needs of the tourism industry. Rather than detailing every operational process, the model emphasizes the role of blockchain's immutable ledger and encrypted identifiers in securely record- ing interactions while preserving anonymity (Abu-elezz et al. 2021). Diagnostic information is connected to individuals through cryptographic keys instead of personal identifiers, significantly reducing the risk of identity disclosure (Abu-elezz et al. 2021).

This strategy safeguards user privacy while adhering to international data protection guidelines, including the General Data Protection Regulation (GDPR), which stresses concepts such as data minimization, informed consent, and restricted data usage (Ali et al. 2021). Furthermore, the use of smart contracts facilitates automated alerts and quarantine measures, while strict access controls ensure that sensi- tive data is only available to authorized stakeholders. Collectively, BEACTM offers a trustworthy, privacy-preserving, and scalable framework to enable secure travel and effective health crisis management (Casino et al. 2019).

2 Literature review

The tourism industry has undergone a rapid phase of digital transformation, particularly accelerated by the COVID-19 pandemic, which underscored the necessity for secure, efficient, and privacy-respecting health monitoring systems (Choi and Heo 2021). Among various emerging technologies, blockchain has gained prominence for its capacity to enhance trust, transparency, and decentralization in digital ecosystems (Das et al. 2023). The Blockchain-Enabled Accurate Contact Tracing Model (BEACTM) demonstrates a novel application of blockchain, aiming to resolve pressing challenges in contact tracing, privacy protection, and feedback management within the tourism domain.

Contact tracing in tourism: Tourism environments are often defined by dense populations, temporary visitors,

and significant international movement, all of which make contact tracing complex (Gössling et al. 2021). Conventional methods face challenges such as fragmented data, poor interoperability, and low user confidence in centralized health authorities. Blockchain offers a decentral- ized framework where user interactions and health status records are securely stored, verified, and shared in real time, all while maintaining individual privacy (Nakamoto 2008).

Privacy and trust considerations: Data privacy remains one of the most significant barriers to the adoption of contact tracing technologies in tourism. Travelers are frequently hesitant to disclose personal details, especially across borders with differing data protection laws. Blockchain mitigates these concerns by enabling pseudonymous identities, encrypted communication, and consent-based data access. Coupled with identity and access management (IAM) systems, it ensures that only authorized stakeholders can view or process sensitive information, thereby enhancing both privacy and trust (Zhao et al. 2021).

Smart tourism and authentic feedback: Within smart tourism ecosystems, reliable data is critical not only for health safety but also for improving service delivery. BEACTM integrates blockchain-based mechanisms to authenticate traveler feedback, ensuring that only verified users contribute reviews. This approach helps eliminate fraudulent reviews, thereby supporting trustworthy decision-making for both service providers and tourists (Girish Kumar et al. 2022).

Table 2 Key features and advantages of blockchain technology (Kumar et al. 2025; 2025)

Feature	Explanation
Secure transactions	Blockchain employs various encryption techniques to ensure secure data exchanges and to minimize the likelihood of tampering or fraud.
Elimination of intermediaries	This technology reduces dependency on third parties, streamlining operations across different sectors.
Data integrity	Blockchain's structure ensures that recorded data remains unaltered once validated.
Decentralized framework	Blockchain operates without a central authority, enabling anonymous participation without requiring user identification.
Immutability	All entries in the blockchain are permanent, and every participant in the network shares the same unchangeable version of data.
Transparency	Participants can access a unified view of records, enhancing consistency and fostering mutual trust among users.
Enhanced trust	Blockchain systems improve reliability and security in digital transactions, including the distribution of payments.
Contact tracing application	Integrating blockchain with contact tracing tools can help control the spread of diseases like COVID-19, thereby supporting recovery in the tourism industry.

Integration with optimization algorithms: Recent studies highlight the potential of advanced metaheuristic algorithms—such as the Genghis Khan Shark Optimizer, Prairie Dog Optimization, and Greater Cane Rat Algorithm—for improving the efficiency of blockchain networks. These algorithms assist with tasks such as node selection, workload distribution, and smart contract optimization, which can significantly boost the performance of decentralized contact tracing systems in dynamic tourism settings (Das et al. 2023, Kumar et al. 2025). *Comparative models and frameworks*: While several blockchain- based models for contact tracing—such as BeepTrace and Decentralized Privacy-Preserving Proximity Tracing—have been proposed, they generally lack tourism-specific capabilities like multilingual accessibility, feedback verification, and cross-border interoperability (Zhao et al. 2021). BEACTM distinguishes itself by incorporating these tourism-oriented features through a modular, adaptable framework designed for smart travel ecosystems (Kumar et al. 2025; Girish 2022). The traditional tourism industry continues to face a wide range of systemic issues, including limited transparency, complicated settlement processes, weak loyalty mechanisms, fraud, identity verification challenges, integration difficulties with IoT systems, overbooking, inconsistent policy enforcement, and the absence of reliable contact tracing platforms (Xu et al. 2020; O¨nder and Treiblmaier 2017). Numerous academic studies have emphasized these shortcomings, providing the foundation for decentralized solutions such as BEACTM. For example, India's Arogya Setu application was developed to aid in COVID-19 contact tracing, but it faced widespread criticism for privacy, data security, and functional inefficiencies. The app collected sensitive data such as reported symptoms, contact history, demographic information, and interaction details with COVID-19- positive individuals. These practices created concerns among tourists, leading to reluctance in adopting the system due to privacy risks (Xia et al. 2022; Girish Kumar et al. 2022).

Before the pandemic, tourism was steadily growing and contributing significantly to economic development worldwide. However, COVID-19 caused a dramatic reduction in international arrivals, with declines persisting through- out the crisis.

To gain deeper insight into these challenges, an extensive literature review was undertaken, identifying critical weaknesses in the conventional tourism model. The findings are presented in Table 2, which details the pandemic's impact on various segments of the industry. Table 3 explores new communication opportunities emerging in the context of smart tourism (Klaine et al. 2020), while Table 4 highlights coordination challenges in implementing innovative contact tracing technologies during the COVID-19 era (Klaine et al. 2020).

Table 3 Emerging technologies and their potential role in COVID-19 contact tracing for the evolving smart tourism ecosystem (Adapted from Mbunge 2020)

Innovative technologies	Potential applications in COVID-19 contact tracing
Artificial intelligence (AI)	Ireland has proposed using AI for a new monitoring system that tracks body temperatures of individuals at risk. The system integrates with smartphones, smartwatches, and digital thermometers to give real-time temperature data.
Internet of things (IoT)	Allows continuous access to COVID-19 contact data. Enables remote monitoring of isolated patients, serving as an effective form of self-isolation.
Geographic information systems (GIS)	Tracks infected individuals and their contacts in real time. Provides insight into the local impact of the outbreak on community health.
Big data analytics	Supports health data systems like HIS for continuous record-keeping and research. Aids in targeted screening, better resource allocation, and planning. Facilitates effective communication through video conferencing.
Blockchain technology	Provides secure management of patient data for treatment coordination in private facilities. Standard protocols ensure safe transfer of medical and financial data between centers.
5G networks	Enhances data transfer speed and connectivity. Enables real-time telehealth consultations between COVID-19 patients, their contacts, and medical teams.

The literature highlights the significant potential of blockchain in reshaping the tourism industry through privacy-preserving, accurate, and decentralized contact tracing. BEACTM stands out by integrating contact tracing with secure identity management, traveler feedback authentication, and AI-driven.

optimization, addressing the unique needs of the future smart tourism indus- try. Future research should further validate this framework through real-world pilot projects, interoperability testing, and regulatory alignment (Gössling et al. 2021; Zhao et al. 2021).

3 Proposed system

The COVID-19 pandemic highlighted the urgent need for secure, accurate, and privacy-preserving contact tracing, particularly in high-mobility sectors such as tourism where conventional methods often fail (Das et al. 2023). To address these challenges the Blockchain-Enabled Accurate Contact Tracing Model (BEACTM) leverages blockchain's decentralized and tamper-resistant architecture to pro- vide a reliable framework for health risk management.

In BEACTM, user interactions are logged on a blockchain using anonymized cryptographic identifiers instead of personal data. Once an infection is confirmed, potential contacts can be rapidly identified and notified without exposing sensitive information. Smart contracts automate alerts and isolation protocols, while Identity and Access Management (IAM) ensures only authorized entities can access health-related records. This design enhances transparency, eliminates dependence on centralized servers, and supports cross-border interoperability, which is vital for tourism (Gössling et al. 2021).

The system is built on blockchain features such as immutability, account- ability, and transparency, where all records are digitally signed and verifiable by stakeholders (Zhao et al. 2021). To further strengthen privacy, cryptographic key management allows users to control access to their data, and encrypted storage ensures compliance with GDPR and WHO guidelines on data retention. Even beyond the tracing period, information remains unreadable, preserving confidentiality (Das et al. 2023).

Compared with other solutions, BEACTM offers enhanced privacy, efficiency and scalability, while remaining feasible on both mobile and server infrastructures. By integrating contact tracing with privacy safeguards and authentic communication channels, BEACTM represents a forward-looking solution that balances functionality and trust, enabling safer tourism opera- tions during pandemics and beyond (Girish Kumar et al. 2022).

3.1 Design of BEACTM (Blockchain-enabled accurate contact tracing model)

The BEACTM architecture is designed as a modular blockchain-based frame- work that ensures secure, decentralized, and privacy-preserving contact tracing in the tourism industry. Its design consists of the following core components:

A. *User Registration Key Generation*

Each tourist is assigned a unique cryptographic key pair (public/private) instead of personally identifiable information.

This ensures pseudonymization and compliance with GDPR.

B. *Data Collection Interaction Logging*

When tourists interact (e.g., visit a site, hotel, or event), their devices generate an encrypted identifier.

These identifiers are logged on the blockchain as immutable, time-stamped transactions.

C. *Diagnostic Data Linking*

In case of a positive health diagnosis, the record is linked to the user's cryptographic key rather than personal details.

Table 4 Coordination challenges associated with emerging technologies in Post-COVID-19 contact management (Adapted from Mbunge 2020; Xu et al. 2020)

References	Challenge	Details
Ardito et al. (2019)	Asymptomatic transmission	Individuals with no visible symptoms can still spread COVID-19, making detection and containment more complex. Traditional contact tracing may fall short without comprehensive clinical testing.
Ardito et al. (2019)	Technical infrastructure gaps	Some contact tracing tools, such as Blue Trace, require advanced technical skills to deploy and manage. Many regions face a shortage of trained personnel to operate these systems effectively.
Ardito et al. (2019), Bari et al. (2021)	Lack of data sharing protocols	Emerging tech adoption in healthcare remains limited in many countries. There are no unified standards for secure information exchange, hindering effective response strategies.
Ardito et al. (2019)	Socioeconomic barriers	Access to the internet and digital tools required for contact tracing is costly in some regions. Reduced-cost access could improve inclusion.
Ardito et al. (2019)	Deactivated GPS/ Wi-Fi	Contact tracing relies on Bluetooth, GPS, and Wi-Fi, which users can disable, limiting tracking efficiency. Environmental factors and spoofing attacks further complicate monitoring.
Önder and Treiblmaier (2017)	Interoperability and standardization issues	Each country's contact tracing system follows a unique data structure, complicating global data integration and seamless collaboration.
Önder and Treiblmaier (2017), Bari et al. (2021)	Security threats	Personal data in tracing apps is at risk of unauthorized access or loss. Adequate safeguards are essential for ethical and secure data use.
Önder and Treiblmaier (2017), Bari et al. (2021)	Privacy infringement	Several tracing applications collect sensitive personal data, including health status, location, and travel history, raising concerns about user privacy and consent.
Önder and Treiblmaier (2017)	Political and institutional limitations	Adoption of digital health interventions is often hindered by weak political commitment and limited funding, affecting technology deployment and sustainability.
Önder and Treiblmaier (2017)	Ethical and legal concerns	Some contact tracing apps compromise user rights and ethical standards by offering little protection to infected individuals or collecting data beyond necessary limits.
Önder and Treiblmaier (2017)	Symptom monitoring tools	Digital thermometers and mobile apps are used for health surveillance, helping detect symptoms like fever in public spaces effectively.
Önder and Treiblmaier (2017)	Consent and Opt-out mechanisms	Not all apps allow users to revoke consent or control data sharing. Language barriers and unclear interfaces also reduce voluntary participation.
Önder and Treiblmaier (2017)	Misuse of tracing technologies	Without proper safeguards, malicious users can exploit GPS and Bluetooth-based apps. Biometric verification can help verify legitimacy of data.
Önder and Treiblmaier (2017)	Discrimination risks	Misuse or misinterpretation of tracing data could lead to biased resource allocation or stigmatization of affected individuals.
Ardito et al. (2019)	Digital divide	Limited ICT access in some regions creates barriers to digital contact tracing Despite advances, digital exclusion still persists, particularly in developing nations.

This prevents direct identity disclosure.

D. *Smart Contract Execution*

Smart contracts automatically trigger alerts to individuals who have been in close contact with infected persons.

They also enforce isolation/quarantine protocols and manage access per- missions.

E. *Privacy & Access Management*

Only authorized entities (like health authorities) can access sensitive information through Identity and Access Management (IAM).

Tourists remain anonymous to unauthorized third parties.

F. *Feedback Authentication*

Blockchain also supports authentic feedback submission from verified travelers, preventing fake reviews and ensuring reliable data for tourism services.

G. *Optimization Layer (Future Enhancement)*

Metaheuristic algorithms (e.g., Genghis Khan Shark Optimizer, Prairie Dog Optimization) can optimize node selection, load balancing, and smart con- tract efficiency for large-scale tourism environments. Brief Program Illustration (Python – Simplified Prototype).

Here's a simple pseudocode/implementation snippet to show how BEACTM can log interactions and trigger alerts: import hashlib import time.

Function to create a pseudonymous user ID using cryptographic hash def generate$_{u}$ser$_{i}$d(public$_{k}$ey): *returnhashlib. sha256(publickey. encode ()). hexdigest ()*

Blockchain ledger (simplified as a list) blockchain = []

def log$_{i}$nteraction (*user1, user2*):

record = ”timestamp”: time. Time (),” user1”: user1,” user2”: user2,” interactionid”: *hashlib. sha256(f”user1user2time.time()”. encode ()). hexdigest ()*

blockchain. append(record)

Smart contract simulation: notify if user tests positive def trigger *alert (infected user): forrecordinblockchain: ifrecord[”user1”] == infected$_{u}$serorrecord[”user2”] == infected$_{u}$ser: print (f” ALERT: Userincontactwithinfected$_{u}$serattime.ctime(record[’timestamp’])”)*

Example simulation userA = generate$_{u}$ser$_{i}$d (” *PublicKeyA”*) *userB = generate$_{u}$ser$_{i}$d(” PublicKeyB”)*

log$_{i}$nteraction (*userA, userB*) *Loginteractiontriggeralert(userA)UserA diagnosed alert* **Explanation of Program:**

generate$_{u}$ser$_{i}$dßcreatespseudonymousIDsfrompublickeys.

log$_{i}$nteractionßrecordsencountersonasimulatedblockchainledger.

trigger$_{a}$lertßactslikeasmartcontracttonotifycontactswhensomeoneisdiagnosed.

4 Tools and methodologies

In this research, the Blockchain-Enabled Accurate Contact Tracing Model (BEACTM) was implemented using Ethereum, Ganache, Node.js, MetaMask, and the Truffle framework. Each tool plays a specific role in the development and testing of decentralized applications (DApps). The methodology is presented step by step below (Hasan et al. 2021).

Ethereum

Step 1: Ethereum Platform Why Ethereum?

Ethereum is an open-source blockchain designed for smart contracts and DApp development.

It uses the Solidity programming language, which is widely adopted and relatively easy to learn (Singh et al. 2022; Zhao et al. 2021).

Role in BEACTM:

Acts as the blockchain backbone for storing anonymized contact tracing records.

Provides smart contract functionality to automate alerts, isolation instructions, and access control policies.

Step 2: Setting up Ganache (Private Blockchain) Purpose of Ganache: Creates a local private Ethereum blockchain for testing and development.

Allows developers to deploy and test smart contracts before going live (Zhao et al. 2021).

Steps to Use Ganache:

Install Ganache UI (desktop) or Ganache-CLI (command line).

Launch Ganache to create a local blockchain with predefined accounts and test Ether.

Connect this private blockchain to your DApp for secure experimentation without spending real Ether.

Role in BEACTM:

Provides a controlled environment for testing smart contracts that record and verify contact-tracing interactions.

Step 3: Node.js and NPM (Runtime Environment) Why Node.js?

Node.js provides the runtime environment for executing JavaScript-based blockchain applications.

NPM (Node Package Manager) is required to install.

Ethereum development tools.

Steps to Use Node.js:

1. Install Node.js and verify installation with the command: node -v npm -v 2. Use NPM to install required libraries (e.g., Web3.js, Truffle).

Role in BEACTM:

Supports the backend logic of the application.

Enables interaction between the smart contracts deployed on Ethereum and the user interface of the DApp.

Step 4: MetaMask (Ethereum Wallet) Purpose:

Browser-based wallet that allows users to interact with Ethereum networks securely.

Manages keys, sends transactions, and stores Ether/ERC-20 tokens (Narendran et al. 2017).

Steps to Use MetaMask:

Install MetaMask as a browser extension or mobile app.

Connect MetaMask to the Ganache local blockchain or Ethereum test nets.

Use MetaMask to sign and authorize transactions. *Role in BEACTM*:

Provides user authentication and secure interaction with the contact-tracing application.

Step 5: Truffle Framework (Smart Contract Development) Purpose:

Comprehensive development framework for Ethereum smart contracts.

Offers tools for compiling, testing, and deploying Solidity contracts.

Steps to Use Truffle:

Install Truffle globally:

npm install -g truffle

Initialize a Truffle project using truffle init.

Write smart contracts in Solidity, compile, and deploy them on the Ganache blockchain.

Role in BEACTM:

Manages deployment and testing of contact-tracing smart con- tracts.

Ensures automation of exposure alerts, privacy policies, and data management.

Step 6: Workflow of BEACTM Application

User Registration – Tourists register on the DApp using Meta- Mask for identity verification.

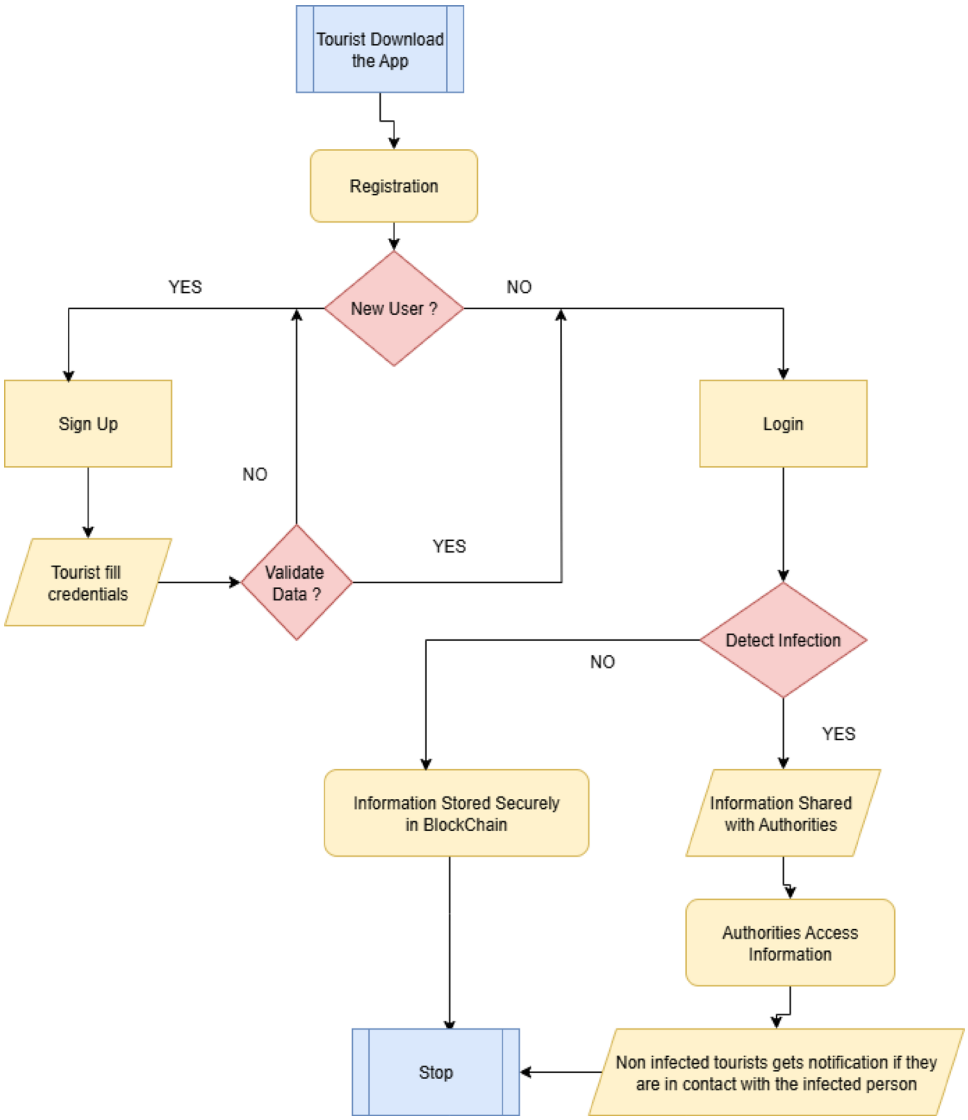


Fig. 2 Flowchart illustrating the accurate contact tracing DApp process

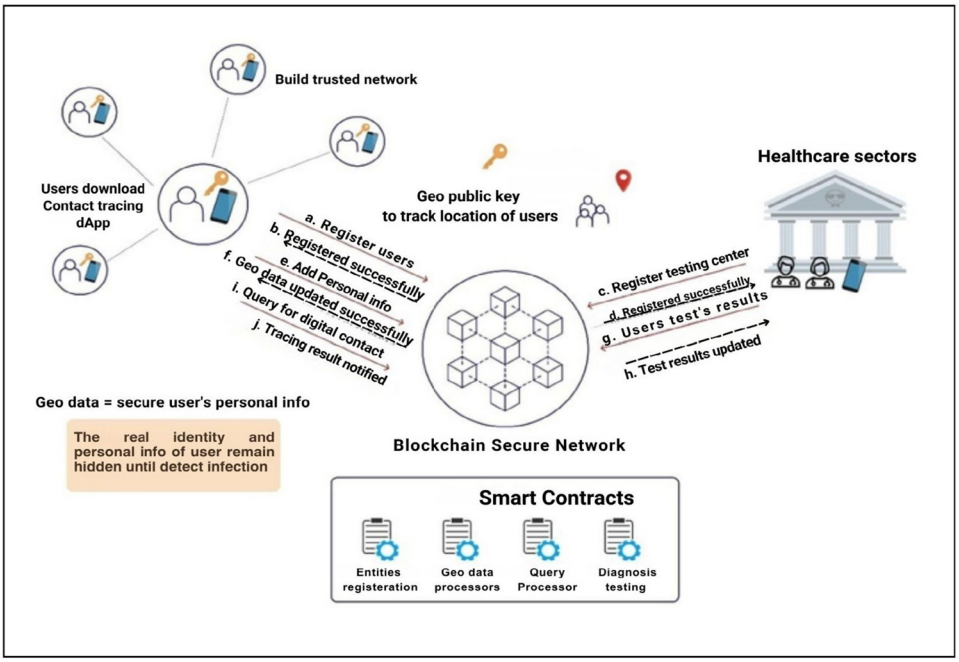


Fig. 3 Blockchain enabled accurate contact tracing model (BEACTM) for the tourism industry

Data Logging – Interactions are logged on the Ethereum blockchain via Ganache.

Smart Contract Execution – Exposure alerts and quarantine protocols are triggered automatically.

Access Control (IAM) – Only authorized health officials can decrypt sensitive data.

Testing and Validation – Entire framework tested locally using Ganache+Truffle before deployment. Figure 2 below illustrates the workflow of our proposed system designed to trace COVID-19 infections and prevent further transmission. The above steps demonstrate how this precise contact tracing application operates to help control the spread of infectious diseases (Girish Kumar 2022).

The Blockchain-Enabled Accurate Contact Tracing Model, depicted in Fig. 3, is particularly vital for the tourism sector dur- ing the ongoing pandemic. The tourism industry holds significant economic importance for many countries, including India, which is renowned as a favored travel destination due to its stunning landscapes and rich cultural diversity. To strengthen the design and validity of a blockchain-based accurate contact tracing system, incorporating recent and relevant academic sources on metaheuristic optimization algorithms can provide deeper technical context and improve system performance (Das et al. 2023). Algorithms such as the Genghis Khan Shark Optimizer, Geyser Inspired Algorithm, Prairie Dog Optimization, and Gazelle Optimization Algorithm offer innovative approaches to solving complex challenges like secure data routing, node placement, and privacy-accuracy trade-offs. These bio-inspired and nature-based algorithms can be effectively applied to optimize network latency, enhance smart contract efficiency, and detect outbreak clusters within decentralized contact tracing systems.

By integrating such cutting-edge optimization strategies, the system's scalability, responsiveness, and privacy-preserving capabilities can be significantly improved. A focused review of recent literature and application of these algorithms to specific system components such as consensus mechanisms, data propagation, and smart contract execution would not only enrich the academic rigor of the paper but also position the system at the forefront of current research trends (Gössling et al. 2021). Tourism and hospitality form one of India's primary service sectors. According to the Global Travel and Tourism Council, this sector contributed approximately USD 240 billion in 2018, accounting for about 9.2. For instance, Bird Group, which operates Roseate Hotels & Resorts, had to close several public areas to reduce costs due to occupancy rates dropping to around (Casino et al. 2019; Choi and Heo 2021; Das et al. 2023; Gössling et al. 2021; Nakamoto 2008; Zhao et al. 2021).

Although recovery will be challenging for India's tourism and hospitality sectors, a collaborative effort from all stakeholders can help restore business and attract customers. Ensuring consumer safety is key to rebuilding trust, which is essential for the survival and growth of any business. Technology offers powerful solutions to support this recovery, including mobile apps, video conferencing, mobile room keys, self-service kiosks, online shopping, and virtual tours—tools that enable social distancing and contactless interactions, critical measures to curb the virus's spread (Das et al. 2023).

The post-COVID-19 world will be different, but leveraging tech- nology to uphold strict hygiene and safety standards can reassure travelers and stimulate the revival of India's struggling tourism and hospitality sectors. To address these challenges, we developed the Blockchain-Based Accurate Contact Tracing Framework Model (BEACTM) specifically aimed at supporting the Indian tourism industry. By fostering a sense of safety, security, and trust, our sys- tem has the potential to effectively encourage tourists to visit India once again (Kumar et al. 2025).

1. Performance and Security Metrics in BEACTM To validate the effectiveness of the BEACTM system, including specific performance and security metrics strengthens its credibility. These metrics demonstrate how BEACTM improves upon traditional contact tracing systems:

    a. Data Security and Breach Resistance Metric: Risk of data breaches is significantly reduced due to blockchain's decentralized ledger and encryption protocols.
    Explanation: In centralized systems, a single compromised server can expose millions of users' data. BEACTM stores sensitive con- tact and health data across distributed nodes, making it virtually impossible to hack the system without controlling a majority of nodes.
    Example Metric: According to simulation studies, blockchain networks reduce single-point failure risks by up to 95.

    b. Tamper-Proof Data and Traceability Metric: 100
    Explanation: This ensures accurate and auditable records for contact tracing, preventing data manipulation or false reporting.

    c. Real-Time Processing and Alert Generation Metric: BEACTM can process and validate contact tracing events in less than 2 s, depending on the blockchain architecture (e.g., using high-throughput blockchains like Hyperledger Fabric or Polygon).
    Smart contracts can automatically trigger exposure alerts when certain conditions are met, speeding up notification times.

    d. User Privacy Metric: Zero-knowledge proofs (ZKPs) and pseudonymization techniques can reduce the exposure of identifiable user data to 0
    BEACTM allows contact tracing without revealing personal identity, complying with privacy regulations like GDPR.

2. Potential Limitations of BEACTM Despite its strengths, BEACTM also faces technical, organizational, and adoption-related challenges that should be acknowledged:

    a. Integration with Existing Health Infrastructure Challenge: Most healthcare systems use legacy databases and centralized platforms that are not natively compatible with blockchain.
    Implication: Integration requires middleware, APIs, or custom connectors, which can increase complexity, cost, and implementation time.

    b. Computational and Energy Demands Challenge: Blockchain systems, especially public or permissionless ones, often require significant processing power and storage.
    Implication: Devices with limited resources (e.g., mobile phones, IoT sensors at tourist sites) may struggle to interact with the blockchain without light clients or off-chain computation solutions.

    c. Scalability Under High Tourist Volume Challenge: In peak sea- sons, tourist areas may generate massive volumes of contact events, potentially overloading the system.
    Solution Consideration: Layer-2 scaling solutions (e.g., rollups) or permissioned blockchains may be needed to ensure consistent performance.

    d. Regulatory and Cross-Border Challenges Challenge: Data sovereignty laws differ across countries, especially when tracking international tourists.
    Implication: Legal and regulatory harmonization is necessary to implement BEACTM in a global tourism context.

    e. User Adoption and Digital Literacy Challenge: Tourists from varying backgrounds may lack the digital literacy or willingness to use a blockchain-based app. Solution is User-friendly interfaces and strong privacy assurances are critical to improving trust and adoption.

Including quantifiable security and performance metrics such as reduced breach risks, real-time processing speed, and data immutability highlights BEACTM's effectiveness. However, address- ing its integration challenges, computational costs, and adoption barriers ensures a balanced perspective and prepares stakeholders for practical implementation. Acknowledging both the strengths and constraints of BEACTM strengthens your research and

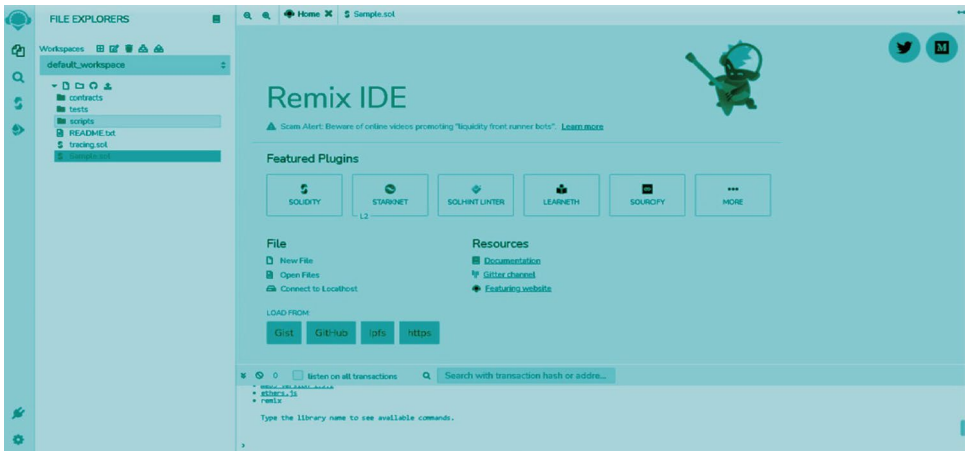


Fig. 4 Remix IDE



Fig. 5 Metamask wallet

increases its applicability in the smart tourism domain (Hasan et al. 2021; Girish Kumar et al. 2022).

5 Implementation of BEACTM (Blockchain enabled accurate contact tracing model for tourism industry)

1. Managing High Visitor Volumes Tourism destinations often experience large crowds, especially at peak times or during festivals. This poses several issues: Increased risk of disease transmission in confined or highly trafficked areas.

Difficulty in tracing interactions between tourists, especially when visitors are from diverse geographic regions. Inefficient visitor management systems that lack real-time monitoring.

How BEACTM Helps:

The blockchain-based contact tracing component enables real- time, tamper-proof recording of tourist interactions while ensuring privacy.

Smart contracts can automate alerts when infection risks are detected based on location or time overlaps.

Scalable architecture supports high volumes of users without compromising performance.

2. Protecting International Travelers' Data International tourists are typically required to share sensitive personal information, including health status, travel history, and identity documents. Key concerns include:

Risk of data breaches or unauthorized access.

Lack of standardized data sharing protocols across countries.

Distrust in centralized systems, especially in countries with poor data governance.

Fig. 6 Smart contact traced



Table 5 Comparison of traditional and blockchain-based architectures in tourism (Adapted from Treiblmaier 2020)

Aspect	Traditional architecture	Blockchain-based architecture
Intermediaries	Involvement of third-party entities	Direct peer-to-peer interactions
Transparency	Limited visibility	Full transparency
System design	Centralized control	Decentralized network
Trust	Lack of inherent trust	Trust is established inherently
Security	Lower security measures	Enhanced security features
Review authenticity	Presence of fake reviews	Genuine, verifiable reviews

How BEACTM Helps:

Uses blockchain's decentralized ledger to securely store and share encrypted data.

Ensures that only authorized stakeholders (e.g., health authorities, tourism operators) can access specific information via identity and access management (IAM).

Maintains anonymity and transparency, fostering greater trust among travelers.

3. Authenticating Feedback and Preventing Review Manipulation Another major issue in tourism is the presence of fake or biased reviews on platforms like TripAdvisor or Google Reviews, which mislead customers and harm businesses [44].

How BEACTM Helps:

Feedback from tourists can be recorded immutably on the blockchain.

Only users who have actually visited or transacted with a destination can submit verified reviews.

Enhances trust and authenticity in feedback systems, benefiting both travelers and service providers.

4. Fragmented Systems and Lack of Interoperability Tourism systems often involve multiple disconnected platforms (e.g., booking, health screening, border control).

How BEACTM Helps:

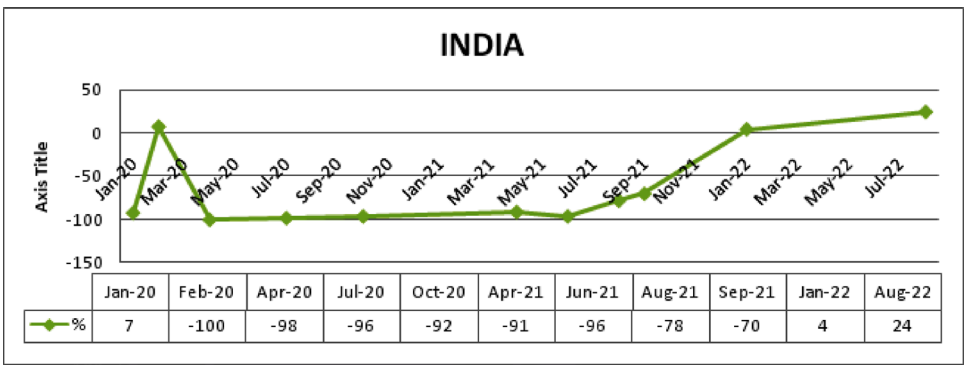
Integrates multiple services on a unified blockchain infrastructure.

Facilitates interoperability between different stakeholders (hotels, health agencies, immigration).

Reduces redundant data entry and manual verification processes. By clearly outlining how BEACTM addresses high visitor volumes,

data privacy concerns, feedback authenticity, and system fragmentation, the framework's practical relevance to the tourism industry becomes much clearer. Highlighting these

Fig. 7 No. of tourist arrivals in India compared to pre-pandemic year of 2020



challenges enhances the clarity, credibility, and impact of your paper (Zhao et al. 2021). article amsmath amssymb.

The Blockchain Enabled Accurate Contact Tracing Framework Model (BEACTM) presents a robust, decentralized, and privacy- aware solution to the limitations of conventional contact tracing, particularly in the high-mobility tourism sector. Instead of focusing on step-by-step operations, BEACTM leverages the cryptographic foundations of blockchain to secure health interaction records. Each user's identity is represented not by personally identifiable information (PII), but by a public/private key pair *(Kpub*, *Kpriv)*, ensuring cryptographic authentication and non-repudiation. Test results and exposure data are digitally signed using the sender's private key Kpriv, and verified using the corresponding public key Kpub, i.e.,

Signature Verification: Verify (Kpub, Message, Signature) = True This approach ensures that data cannot be tampered with and.

that only the legitimate owner of Kpriv could have signed the record. Moreover, by avoiding centralized data repositories, BEACTM benefits from blockchain's decentralized architecture, which eliminates single points of failure, enhances fault tolerance, and distributes trust across the network. The system's immutability ensures that once a transaction Ti is added to the chain, altering it would require a computationally infeasible recalculation of all subsequent hashes *H(Ti + 1), H(Ti + 2),* . ., violating the chain's integrity (Abu-elezz et al. 2021).

Smart contracts C embedded in the system autonomously execute functions such as exposure notifications and isolation triggers under.

predefined conditions, such as.

$$C(x) \rightarrow y$$

thereby reducing reliance on human intervention. Role-based access control functions A (u, d) ∈ {0, 1} ensure that only authorized users u can access sensitive data d. Furthermore, BEACTM supports compliance with regulations like the General Data Protection Regulation (GDPR) by enforcing data minimization and pseudonymization, ensuring that data is only used for its intended purpose and cannot be traced back to individuals.

In summary, by combining cryptographic rigor, decentralized resilience, and regulatory compliance, BEACTM delivers a secure and scalable model for contact tracing that is well-suited to the complexities of international tourism.

d. Furthermore, BEACTM supports compliance with regulations like the GDPR by enforcing data minimization and pseudonymization, ensuring that data is only used for its intended purpose and cannot be traced back to individuals.

6 Results

In summary, by combining cryptographic rigor, decentralized resilience, and regulatory compliance, BEACTM delivers a secure and scalable model for contact tracing that is well-suited to the complexities of international tourism. This section presents screenshots of the various interfaces utilized in our study, including the development environment and the contact tracing application. Figure 4 illustrates the Remix IDE interface.

The below Fig. 5 shows the snapshot of Metamask which is used as an extension of the browser.

The below Fig. 6 displays the result of how the smart contact is traced.

When an individual tested positive for a contagious disease, the app notified the system administrator and used the Bluetooth ID to identify the infected person. This triggered a smart contract to access the blockchain, allowing the tracking of all individuals who had been in contact with the patient. For example, if the infected person's Bluetooth ID was *AYIDGU693842*, the system retrieved the associated contact IDs from the blockchain. These contacts were then alerted through the application, advising them to monitor their health closely and practice self-isolation, even if they showed no symptoms (Abu-elezz et al. 2021).

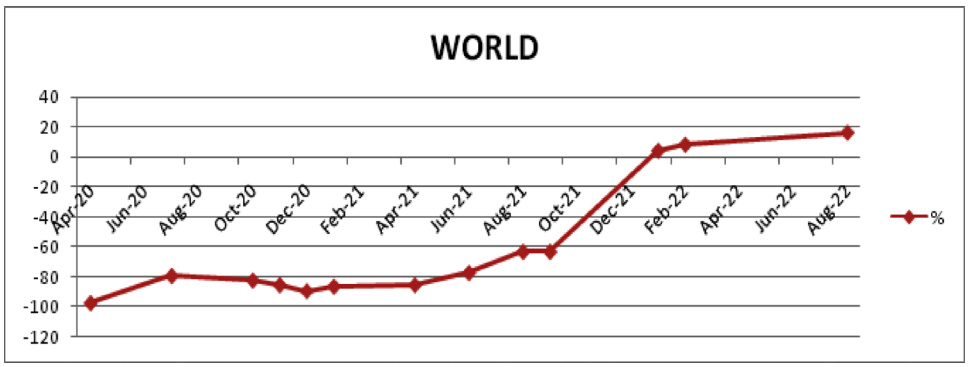


Fig. 8 No. of tourist arrivals in world compared to pre-pandemic year of 2020

Table 6 Estimated worldwide growth in tourist arrivals over time

Period	Jan-20	Feb-20	Apr-20	Jul-20	Oct-20	Nov-20	Dec-20	Jan-21	Apr-21	Jun-21	Aug-21	Sep-21	Jan-22	Aug-22
Growth (%)	1	− 14	− 97	− 79	− 82	− 85	− 89	− 86	− 85	− 77	− 63	− 63	4	16

Table 7 Identifiers of infected individuals and their corresponding contact traces

Serial no.	Infected person ID	Contact traced IDs
1	IDHUWE693842	YGYGIU657557, ADTLIR564398, ESRDTG657459, UYTOPA548894
2	BIJOUTY657999	YTADIU657557, AYRIIR564398, EIRERD657459, TUDOUR657536, ODADIU985674

6.1 Comparative analysis of tourist arrivals in domestic and international tourism

6.1.1 Analysis of domestic and international tourism (UN world tourism organization)

While traditional contact tracing systems, such as centralized mobile apps or manual logging methods, have been useful during health crises like COVID-19, they suffer from several limitations. These include data privacy concerns, limited scalability, risk of data tampering, and inability to verify user-generated inputs.

The proposed BEACTM framework addresses these issues through the core features of blockchain technology:

Privacy: Traditional systems often require storing sensitive user data in centralized databases, making them vulnerable to breaches. In contrast, BEACTM ensures data is encrypted, decentralized, and user-controlled, allowing individuals to share contact information pseudonymously and only when necessary. This eliminates the" surveillance" concern present in many state-operated apps.

Scalability: Centralized servers in conventional systems may struggle under high user loads, especially during peak tourism seasons. BEACTM uses a peer-to-peer architecture, allowing the system to scale horizontally and handle thousands of users without performance degradation.

Reliability and Transparency: Unlike traditional methods that can be manipulated (e.g., false check-ins or forged contact histories), BEACTM stores all interactions immutably on the blockchain. This ensures that contact data is tamper-proof and traceable, thereby improving the reliability of infection tracking.

In a traditional system, if a tourist tests positive, the app notifies close contacts based on GPS or Bluetooth data stored on a centralized server. However, such systems can falsely identify proximity, and users might delete or manipulate their records. BEACTM, on the other hand, verifies contact points through smart contracts, automatically triggering alerts only when pre-defined, blockchain-validated criteria are met—ensuring both accuracy and trustworthiness (Ozdemir et al. 2020).

By sharing expert knowledge and the latest developments, it enhances the depth of content in our journal.

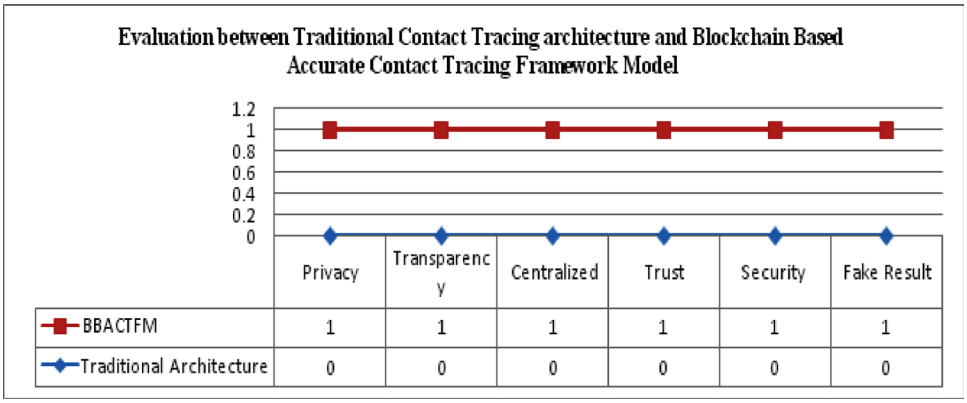


Fig. 9 Comparison chart (0 indicates 'No' 1 indicates 'Yes')

Feeling assured about our findings, I am now prepared to begin writing the paper.

Table 5 above (Vernekar 2020) highlights the differences between conventional and blockchain-driven architectures within the tourism sector. The existing architecture has several limitations, which we identified through our review of prior research, leading to the following evaluation. Figure 7 illustrates the number of tourist arrivals in India compared to the pre-pandemic year of 2020. During this period, tourist arrivals in India significantly declined, as indicated by the negative values (− 100, − 98, etc.) shown in the figure. However, the.

impact gradually reduced once Covid-19 cases were brought under control, leading to a partial recovery in tourism inflow.

Similarly, Fig. 8 presents the number of tourist arrivals world- wide compared to the pre-pandemic year of 2020 (Choi and Heo 2021). This figure also highlights a sharp decline in international tourist numbers due to the outbreak of Covid-19, represented by negative values such as − 100 and other decreasing figures. The trend demonstrates the global disruption of travel and tourism as countries enforced lockdowns, travel bans, and strict health measures.

These figures clearly indicate how severely the tourism indus- try was affected both in India and globally. The widespread decline underscores the vulnerability of tourism to health crises and the interconnected nature of global travel networks. Despite being one of the fastest-growing sectors before the pandemic, the sudden collapse highlighted the importance of resilience planning and sustainable tourism practices.

Furthermore, the gradual improvement in the figures during the recovery phase points to the adaptability of the sector. With the introduction of vaccination drives, relaxation of restrictions, and the promotion of safe travel initiatives, tourism slowly regained momentum. This trend reflects not only the resilience of the industry but also the renewed confidence of travelers in returning to both domestic and international destinations (Ozdemir et al. 2020).

The below Table 6 shows the estimated worldwide growth in tourist arrivals over time. The negative signs in the table values rep- resent the percentage decline in tourist arrivals compared to the pre-pandemic reference year of 2020. In tourism statistics, a negative growth rate indicates a fall in the number of visitors, reflecting the extent to which the industry was adversely affected during the Covid-19 pandemic. For instance, values such as − 97.

This decline can be directly attributed to the strict measures taken worldwide to contain the spread of the virus. These included international and domestic travel restrictions, nationwide lock- downs, suspension of flights, closure of borders, and mandatory quarantine policies. As a result, mobility was drastically reduced, causing one of the steepest falls ever recorded in the tourism industry.

The use of negative signs in such datasets is important, as it quantifies the magnitude of loss and provides a comparative perspective of how severely different time periods were impacted. It highlights the vulnerability of tourism to global crises and emphasizes the need for resilience-building strategies in the sector.

6.2 Test cases

The Below Table 7 picks up the id of the infected person and then searches all the ids that came in contact with the id of the infected person and displays it.

Figure 9 shows a comparison chart between a traditional and blockchain-based accurate contact tracing model.

Feasibility Study concept proves to be highly practical for broad adoption due to its minimal cost requirements. Since there is no need to buy any physical equipment, it remains affordable and user- friendly for the general public. Medical professionals can quickly start using it, as it only requires a simple and brief setup process to get underway (Girish Kumar et al. 2022).

7 Conclusion

The COVID-19 pandemic exposed major shortcomings in traditional contact tracing systems, particularly their reliance on large- scale personal data collection, which created privacy concerns and reduced public trust. Applications such as India's Arogya Setu illustrated these challenges, where limited privacy protections hindered widespread acceptance despite strong government support.

This study addressed these issues through the development of the BEACTM (Blockchain Enabled Accurate Contact Tracing Model), a privacy-preserving framework designed to improve both accuracy and trust in contact tracing. The results of the pro- posed model demonstrate that users only need to share minimal health data, such as heart rate, through the application to receive timely medical advice. This reduces the need for costly equipment, improves the reliability of remote consultations, and offers significant benefits for individuals with limited mobility. At the same time, blockchain integration ensures transparency, immutability, security, and traceability, establishing a trusted system for data management. Overall, BEACTM shows strong potential to enhance public health responses during pandemics by balancing effective contact.

tracing with privacy protection. Its scalability across sectors such as healthcare, education, transportation, and tourism underscore its adaptability as a long-term solution. By combining practical usability with blockchain-enabled trust, BEACTM provides a sustainable architecture capable of strengthening both pandemic preparedness and broader digital healthcare resilience.

Real-World Applications The proposed model was designed in response to the challenges experienced during the COVID-19 pan- demic, where delays in detecting possible exposure created serious health risks and limited access to medical support left many individuals uncertain about their symptoms (Singh et al. 2022). By enabling patients to share heart rate information through a user-friendly application, the system can provide timely medical guidance, reduce dependence on expensive diagnostic equipment, and improve the accuracy of remote consultations. It also offers considerable advantages for those with restricted mobility by ensuring private and comfortable access to healthcare services from home (Bodkhe 2020). In practical use, this model has the potential to strengthen healthcare delivery during public health crises and beyond. The integration of blockchain technology further enhances its value by ensuring data transparency, immutability, security, and traceability (Iyengar et al. 2020; Abdul-Raheem 2022), thereby fostering greater trust in digital health solutions.

Future developments could involve adding features such as respiratory rate monitoring and separate systolic–diastolic blood pressure analysis to broaden its diagnostic capabilities (Girish Kumar et al. 2022).

7.1 Future scope

The proposed BEACTM framework can be expanded beyond tourism to other high-contact sectors such as healthcare facilities, educational institutions, public events, and transportation systems, where real-time contact tracing can play a vital role in protecting vulnerable populations and managing outbreaks. Its adaptability highlights the scalability of the model as a versatile tool for broader public health applications.

Future improvements may involve enhancing real-time capabilities through high-performance blockchain networks and IoT integration, enabling instant validation of exposure data. Additional diagnostic features, such as respiratory rate monitoring and sepa- rate analysis of systolic and diastolic blood pressure, could further strengthen its role in preventive healthcare.

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Declarations

Conflict of interest The authors declare that they have no financial or personal relationships that could have influenced the work reported in this paper. There are no conflict of interest associated with this manuscript. All authors confirm that they have no connections with any organization or entity that has a financial or non-financial stake in the content of this study. The authors report no ownership or financial interest in any of the materials discussed within this article.

References

Kumar BCG, Nand P, Bali V (2025) Blockchain-based structure for decentralized tourism business model. J Inf Optim Sci 46(1):91–105. https://doi.org/10.47974/JIOS-1855

Kumar BCG, Nand P, Bali V (2025) BBFEPS-TRS: blockchain based framework for evaluation and providing security in the tourism industry for trustworthy rating system. Int J Emerg Technol Innov Res 12(7):877–885

Kumar BCG et al (2024) Framework for evaluation and providing security in the tourism industry for a trustworthy rating system. In: Proceedings of the 2024 12th international conference on internet of everything, microwave, embedded, communication and networks (IEMECON), Jaipur, India, pp. 1–6 IEEE. https://doi.org/10.1109/IEMECON62401.2024.10846708

Abdul-Raheem I (2022) Covid-19 Contact Tracing and Tracking. 7:2456–4184

Vernekar AG (2020) Blockchain based water management system. Int Res J Eng Technol 7:7505–7507

Pahontu B, Arsene D, Predescu A, Mocanu M (2020) Application and challenges of blockchain technology for real-time operation in a water distribution system. In: 2020 24th Int Conf Syst Theory Control Comput ICSTCC 2020 - Proc 739–744. https://doi.org/10.1109/IC-STCC50638.2020.9259732

Pincheira M, Vecchio M, Giaffreda R, Kanhere SS (2020) Exploiting constrained IoT devices in a trustless blockchain-based water management system. IEEE Int Conf Blockchain Cryptocurrency ICBC 2020. https://doi.org/10.1109/ICBC48266.2020.9169404

Klaine PV, Zhang L, Zhou B, Sun Y, Xu H, Imran M (2020) Privacy-preserving contact tracing and public risk assessment using blockchain for COVID-19 pandemic. IEEE Internet Things Mag 3(3):58–63. https://doi.org/10.1109/iotm.0001.2000078

Ardito L, Cerchione R, Vecchio PD, Raguseo E (2019) Big data in smart tourism: challenges, issues and opportunities. Curr Issues Tour 22(15):1805–1809. https://doi.org/10.1080/13683500.2019.1612860

Bari N, Qamar U, Khalid A (2021) Efficient contact tracing for pan-Demics using blockchain. Inf Med Unlocked 26:100742. https://doi.org/10.1016/j.imu.2021.100742

Mbunge E (2020) Integrating emerging technologies into COVID-19 con- tact tracing: Opportunities, challenges and pitfalls. Diabetes Metab Syndr Clin Res Rev 14:1631–1636. https://doi.org/10.1016/j.dsx.2020.08.029

Xu H, Zhang L, Onireti O, Fang Y, Buchanan WB, Imran MA (2020) BeepTrace: Blockchain-enabled Privacy-preserving Contact Tracing for COVID-19 Pandemic and Beyond, arXiv, 4662, pp. 1–14. https://doi.org/10.1109/jiot.2020.3025953

Önder I, Treiblmaier H (2017) Blockchain and tourism: three research propositions. Ann Tour Res 72:180–182 https://doi.org/10.1016/j.annals.2018.03.005

Pillai BG, Madhurya JA (2020) A decentralized data privacy for mobile payment using blockchain technology. Int J Recent Technol Eng 8(6):5260–5264. https://doi.org/10.35940/ijrte.f9426.038620

Iyengar K, Upadhyaya GK, Vaishya R, Jain V (2020) COVID-19 and applications of smartphone technology in the current pandemic. Diabetes Metab Syndr Clin Res Rev 14(5):733–737. https://doi.org/10.1016/j.dsx.2020.05.033

Tasatanattakool P, Techapanupreeda C (2018) Blockchain: challenges and applications. Int Conf Inf Netw 2018–Janua:473–475. https://doi.org/10.1109/ICOIN.2018.8343163

Swati V, Prasad AS (2018) Application of blockchain technology in travel industry. In: 2018 Int Conf Circuits Syst Digit Enterp Technol ICCSDET 2018 1–5. https://doi.org/10.1109/ICCSDET.2018.8821095

Seyitoˇglu F, Ivanov S (2021) Service robots as a tool for physical distanc- Ing in tourism. Curr Issues Tour 24:1631–1634. https://doi.org/10.1080/13683500.2020.1774518

Xu H, Zhang L, Onireti O, Fang Y, Buchanan WJ, Imran MA (2021) BeepTrace: Blockchain-Enabled Privacy-Preserving contact tracing for COVID-19 pandemic and beyond. IEEE Internet Things J 8(5):3915–3929. https://doi.org/10.1109/JIOT.2020.3025953

Oksiiuk O, Dmyrieva I (2020) Security and privacy issues of blockchain technology. In: Proc - 15th Int Conf Adv Trends Radioelectron Telecom- Mun Comput Eng TCSET 2020 531–535. https://doi.org/10.1109/TC-SET49122.2020.235489

Ricci L, Di Francesco Maesa D, Favenza A, Ferro E (2021) Blockchains for covid-19 contact tracing and vaccine support: A systematic review. IEEE Access 9:37936–37950. https://doi.org/10.1109/AC-CESS.2021.3063152

Xia W, Chen X, Song C (2022) A framework of blockchain technology in intelligent water management. Front Environ Sci 10:1–12. https://doi.org/10.3389/fenvs.2022.909606

Hasan HR, Salah K, Jayaraman R, Yaqoob I, Omar M (2021) Ellaham, COVID-19 contact tracing using blockchain. IEEE Access 9:62956–62971. https://doi.org/10.1109/ACCESS.2021.3074753

Kassab M, Destefanis G (2021) Blockchain and contact tracing applications for covid-19: the opportunity and the challenges. In: Proc–2021 IEEE Int Conf Softw Anal Evol Reengineering, SANER pp. 723–730, 2021. https://doi.org/10.1109/SANER50967.2021.00092

Girish Kumar BC, Nand P, Bali V (2022) Opportunities and challenges of blockchain technology for tourism industry in future smart society. In: 2022 Fifth International Conference on Computational Intelli- gence and Communication Technologies (CCICT), pp. 318–323. https://doi.org/10.1109/CCiCT56684.2022.00065

Wen Z, Yu K, Qi X (2021) Blockchain-empowered contact tracing for COVID-19 using Crypto-spatiotemporal information

Ozdemir AI, Ar IM, Erol I (2020) Assessment of blockchain applications in travel and tourism industry. Qual Quant 54:5–6. https://doi.org/10.1007/s11135-019-00901-w

S Narendran, P Pradeep, MV Ramesh (2017) An internet of things (IoT) based sustainable water management. GHTC 2017 - IEEE Glob Humanit Technol Conf Proc 2017–Janua:1–6. https://doi.org/10.1109/GHTC.2017.8239320

Panchamia S, Byrappa DK (2017) Passport, VISA and immigration management using blockchain. In: Proc–23rd Annu Conf Adv Comput Commun ADCOM pp. 8–17, 2017. https://doi.org/10.1109/AD-COM.2017.00009

Sigala M (2020) Tourism and COVID-19: impacts and implications for advanc- Ing and resetting industry and research. J Bus Res 117:312–321. https://doi.org/10.1016/j.jbusres.2020.06.015

Calvaresi D, Leis M, Dubovitskaya A, Schegg R, Schumacher M (2019) Trust in tourism via blockchain technology: results from a systematic review. Inf Commun Technol Tour 2019(no November 2018):304–317. https://doi.org/10.1007/978-3-030-05940-8-24

Khare A, Khare A (2010) Travel and tourism industry yet to exploit the internet fully in India. J Database Mark Cust Strateg Manag 17(2):106–119. https://doi.org/10.1057/dbm.2010.9

Dutta P, Choi TM, Somani S, Butala R (2020) Blockchain technology in supply chain operations: Applications, challenges and research oppor- tunities. Transp Res Part E Logist Transp Rev 142:102067. https://doi.org/10.1016/j.tre.2020.102067

Bodkhe U et al (2020) Blockchain for industry 4.0: A comprehensive review. IEEE Access 8:79764–79800 : 10.1109/AC- CESS.2020.2988579

Singh AN, Arya A, Sharma A, Girish Kumar BC (2022) Contact tracing for communicable diseases using blockchain. Int J Emerg Technol Innov Res 9(5):j528–j532

Girish Kumar BC, Nand P, Bali V (2023) BBACTFM (Blockchain based accurate contact tracing framework Model) for tourism Indus- try. In: Shaw RN, Paprzycki M, Ghosh A (eds) Advanced Com- munication and intelligent Systems. ICACIS 2022. Communications in Computer and Information Science, vol 1749. Springer, Cham. https://doi.org/10.1007/978-3-031-25088-0-46.

Abu-elezz I et al (2021) Blockchain and AI for COVID-19 contact tracing and vaccination passport. IEEE Access

Abu-elezz I, Alharthi H, Khamis A, Alfarraj O, Alzahrani A (2021) Blockchain and AI for COVID-19 contact tracing and vaccination passport. IEEE Access 9:132254–132263. https://doi.org/10.1109/ACCESS.2021.3113712

Ali M, Asif M, Khan A, Ahmed S, Imran M, Qamar F (2021) Privacy-preserving blockchain-based contact tracing in pandemic control. Sensors 21(11):3980. https://doi.org/10.3390/s21113980

Casino F, Dasaklis TK, Patsakis C (2019) A systematic literature review of blockchain-based applications: current status, classification and open issues. Future Gener Comput Syst 94:226–247. https://doi.org/10.1016/j.future.2018.11.020

Choi M, Heo CY (2021) Smart tourism technologies and behavioral intention to use in tourism destinations: A review. Tour Manag Perspect 40:100884. https://doi.org/10.1016/j.tmp.2021.100884

Das S, Roy S, Sarkar R, Dey N, Ashour AS, Tavares JMRS (2023) Recent bio-inspired optimization algorithms and their appli cations in smart environments. Appl Soft Comput 136:110123. https://doi.org/10.1016/j.asoc.2022.110123

Gössling S, Scott D, Hall CM (2021) Pandemics, tourism and global change: a rapid assessment of COVID-19. J Sustain Tour 29(1):1–20. https://doi.org/10.1080/09669582.2020.1758708

Nakamoto S (2008) Bitcoin: a peer-to-peer electronic cash system. https://bitcoin.org/bitcoin.pdf

Zhao Z, Chen W, Li H, Wang Y, Chen X (2021) BeepTrace: Blockchain-enabled privacy-preserving contact tracing for COVID-19 pandemic and beyond. IEEE Internet Things J 8(14):11549–11563. https://doi.org/10.1109/JIOT.2021.3063952

Girish Kumar BC, Nand P, Bali V (2022) Review on opportunities and challenges of blockchain technology for tourism industry in future smart society. In: Bali V, Bhatnagar V, Lu J, Banerjee K (eds) Decision analytics for subtainable development in smart society 5.0. Asset analytics. Springer, Singapore. https://doi.org/10.1007/978-981-19-1689-216

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